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**A Multi-Agent Generative AI Platform for Multi-Morbidity
Conflict Resolution and Medication–Diet Interaction
Verification**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA6502 – GENERATIVE AI AND LARGE SCALE MODELS

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI)

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “A Multi-Agent Generative AI Platform for Multi-Morbidity Conflict Resolution and Medication–Diet Interaction Verification” has been carried out by N Goutham(Reg. No. 192472024) under the supervision of Dr. Nithisha. J and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering (AI) programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I am N Goutham, of the Department of Computer Science and Engineering (AI), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that this individual capstone report on “A Multi-Agent Generative AI Platform for Multi-Morbidity Conflict Resolution and Medication–Diet Interaction Verification” (the MediSync AI platform) documents work carried out as part of a four-member project team, and presents the project in my own words with particular emphasis on Module 4 – AI Decision Support, Report Generation & Healthcare Dashboard – which was my own primary area of design and development, as recorded in the Individual Contribution Statement of this report. To the best of my knowledge, the content presented herein is accurate and has been prepared in accordance with the principles of engineering ethics and academic integrity, with all sources, references, data and materials appropriately acknowledged. I further declare that the data, results, analysis and findings presented have not been fabricated, falsified or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional and professional requirements have been followed throughout the planning, execution, analysis and documentation of this report.

Place: Chennai

Date:

Signature of the Student with Name

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Individual Contribution Statement

(MediSync AI was built by a four-member team. This report is Goutham N's individual capstone submission on the shared project, written to document the complete system while foregrounding the module he personally designed and implemented.)

Note: the remaining three modules of the shared MediSync AI project – patient profile management, multi-agent medication verification, and AI decision support & report generation – were contributed by teammates respectively. Hemanth P, N. Chaitanya Ram and M. Mohamed Bazith respectively, and are described in this report for project completeness and context.

Student Name / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Individual Contribution (%)
N. Goutham 192472024	Module 3 – Medication–Diet Interaction Verification & Personalized Nutrition	Implemented the Medication–Diet Interaction Verification and Personalized Nutrition module, developed dynamic food recommendations based on patient diseases, medications, allergies, and dietary preferences, and integrated personalized nutrition guidance.	Tested medication–food interactions and personalized nutrition recommendations across different disease and medication combinations	Authored this individual report in full, covering the whole project with emphasis on Module 3	25% of overall team project; 100% of individual module contribution

ABSTRACT

Patients managing more than one chronic illness at the same time — for instance diabetes, high blood pressure and kidney disease together — are usually placed on several medicines at once, a pattern clinicians refer to as polypharmacy. Studies on multimorbidity and polypharmacy report widely varying but frequently high rates across both adult and elderly populations, and they consistently show that taking a larger number of medicines together raises the risk of harmful drug–drug, drug–disease and drug–food interactions. Today, this cross-checking is done manually by clinicians using separate reference tools, which is slow, dependent on individual experience, and rarely connected to dietary advice — so a conflict between a prescribed medicine and a patient's diet can easily be missed. This project addresses the engineering problem of building a fast, unified, personalised way to check for medication and dietary conflicts in multi-morbidity patients.

MediSync AI is proposed as the solution: a multi-agent generative-AI healthcare decision-support platform built on the CrewAI orchestration framework. Five specialised, cooperating agents — Disease Analysis, Medication Verification, Interaction Detection, Diet & Nutrition, and Report Generation — process a patient's profile in stages. Each agent's output is anchored to real reference material through Retrieval-Augmented Generation (RAG): a ChromaDB vector store indexes drug data (OpenFDA/RxNorm), nutrition data (USDA FoodData Central) and clinical guideline text, and the retrieved passages steer a Google Gemini 2.5 Flash language model that produces the final personalised health summary, medication alerts and diet recommendations. Patient and clinical data are stored in a PostgreSQL database, with results shown through a web dashboard. As part of the four-member development team, my own primary responsibility was Module 4: building the Report Generation agent's RAG-retrieval pipeline and Gemini-based summarisation, and building the dashboard that presents the combined output of all five agents.

Keywords: Multi-Agent Systems; Generative AI; Retrieval-Augmented Generation (RAG); Drug–Drug Interaction; Medication–Diet Interaction; Large Language Models; Healthcare Decision Support

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
AI	Artificial Intelligence
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
DDI	Drug–Drug Interaction
CDSS	Clinical Decision Support System
API	Application Programming Interface
DB	Database
RxNorm / OpenFDA	US drug nomenclature / regulatory drug data sources
USDA	United States Department of Agriculture (FoodData Central)
DPDP Act	Digital Personal Data Protection Act, 2023 (India)
UI	User Interface
mg	Milligram (dosage unit)

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Multimorbidity — having two or more chronic conditions at once — and the polypharmacy it drives are well documented in clinical literature. Systematic reviews report multimorbidity prevalence ranging anywhere from under 5% to over 90%, and polypharmacy prevalence from roughly 3% to over 85%, depending heavily on the population studied and how the terms are defined, with rates consistently climbing with age. A patient juggling diabetes, hypertension and kidney disease at the same time is typically prescribed medicines from several different drug classes together, each carrying its own dosage schedule, contraindications and dietary precautions. Meanwhile, recent progress in generative AI — large language models combined with retrieval-augmented generation and multi-agent orchestration frameworks such as CrewAI — has made it realistic to break a complex reasoning task like cross-checking a patient's full medical and medication history into several specialised, cooperating AI agents rather than one large model call. Our team's project, MediSync AI, applies this architectural approach to the specific problem of multi-morbidity medication and medication–diet conflict resolution; my own contribution centred on Module 4, which turns the outputs of the other agents into a consolidated, retrieval-grounded report and dashboard.

1.2 Need for the Project

- Why it matters: manually cross-checking several prescriptions against several diagnoses and dietary habits is slow and depends on individual clinical experience, raising the chance that a harmful interaction slips through.
- Who is affected: multi-morbidity and chronic-disease patients on several concurrent medicines, along with the physicians and pharmacists who have to verify their safety.
- Existing gaps: conventional drug-interaction reference tools work in isolation from nutrition guidance, so a medication–food conflict (for example, a statin and grapefruit juice) often never reaches the patient alongside their medication advice.
- Engineering significance: a multi-agent, RAG-grounded generative-AI pipeline can bring disease analysis, medication verification, interaction detection and personalised nutrition guidance together into one explainable, patient-specific report — which is exactly what the Module 4 report-generation and dashboard layer I built was designed to deliver.

1.3 Problem Statement

Given a patient profile made up of multiple diagnosed chronic conditions, a list of currently prescribed medicines with dosages, and stated dietary preferences, design a multi-agent generative-AI system that (i) analyses the combination of diseases for known conflicts, (ii) verifies each medicine's dosage and detects drug–drug and drug–disease interactions, (iii) detects medication–food interactions and produces a personalised, categorised diet recommendation, and (iv) consolidates all of this into a single, clinician-readable report — while grounding every generated statement in retrieved reference data (drug databases, nutrition databases and medical

guidelines) to reduce the risk of ungrounded or hallucinated LLM output in a healthcare-safety-sensitive setting. This last consolidation-and-grounding step is where my own module, Module 4, sits within the overall pipeline.

1.4 Project Aim

To contribute to, and individually document, MediSync AI — a multi-agent generative-AI healthcare decision-support platform that detects multi-morbidity medication conflicts and medication–diet interaction risks and generates personalised, RAG-grounded recommendations to support, not replace, clinical decision-making — with particular focus on the report-generation and dashboard layer that consolidates the platform's output.

1.5 Project Objectives

1. To detect conflicts among multiple co-occurring chronic diseases accurately and efficiently.
2. To verify drug–drug and drug–disease interactions automatically using trusted medical knowledge databases.
3. To identify medication and dietary incompatibilities for improved patient safety.
4. To generate personalised meal recommendations based on the patient's individual medical conditions and preferences.
5. To consolidate the outputs of the disease-analysis, medication, and diet agents into a single, retrieval-grounded, decision-support report and dashboard (my own primary objective, Module 4).
6. To improve treatment quality while reducing adverse-interaction risk through a consolidated, generated healthcare report.

1.6 Scope

Included:

- A functional web-based prototype covering patient health-profile intake, multi-agent medication-conflict detection, medication–diet interaction verification with personalised nutrition guidance, and an AI-generated report/dashboard.
- Within the shared project, my individual scope covers the RAG-based Report Generation agent and the healthcare dashboard that surfaces every other agent's output (Module 4).

Excluded:

- Integration with hospital electronic health record (EHR) systems, formal regulatory/clinical validation as a certified medical device, a native mobile application, and wearable/IoT-based real-time monitoring — identified as future work in Chapter 5.

Assumptions:

- Patient-entered demographic, condition and medication data is assumed accurate and complete at intake; each session corresponds to a single patient profile; the third-party drug, nutrition and LLM APIs used are assumed available and current.

Boundary conditions:

- MediSync AI's output is decision-support information meant to assist, not replace, a qualified physician's clinical judgement; the prototype has been functionally tested on representative profiles and has not undergone formal clinical trial validation.

1.7 Expected Engineering Outcome

A validated, functional multi-agent generative-AI web prototype (MediSync AI) that ingests a multi-morbidity patient's health and medication profile and produces medication-alert, interaction-detection and personalised-nutrition outputs through a consolidated, RAG-grounded, LLM-generated healthcare report and dashboard — the consolidation layer being my own individual deliverable within the team's project.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and technical documentation were located through keyword searches ("multi-agent LLM healthcare", "retrieval-augmented generation clinical decision support", "drug–drug interaction detection AI", "polypharmacy management systems", "CrewAI multi-agent framework") across academic sources (arXiv, PubMed/PMC, Springer), together with primary technical documentation for the RAG framework, agentic orchestration frameworks, and the public drug/nutrition data sources used by the project.

2.2 Review of Existing Work

Rule-based clinical decision support systems: established commercial drug-interaction checkers rely on curated, rule-based interaction databases maintained by pharmacology experts. They are accurate for known, catalogued interactions, but are static, need manual updates, and typically do not fold in personalised dietary guidance alongside the medication checks.

Retrieval-Augmented Generation: Lewis et al. proposed retrieval-augmented generation, a framework in which a neural retriever supplies relevant passages from an external corpus to condition a generative model's output; later biomedical work has shown that grounding LLM output in retrieved evidence reduces hallucination and gives more current, traceable knowledge than relying on a language model's parametric memory alone — which is the direct motivation for the ChromaDB-backed RAG layer that sits underneath every agent in this project, including the Report Generation agent I built.

Single-agent LLM medical assistants: general-purpose medical chatbots built on one LLM call give flexible natural-language interaction, but without dedicated retrieval grounding or task decomposition they are more prone to hallucinated or inconsistent recommendations, and typically lack an explicit, auditable interaction-verification step.

Agentic / multi-agent LLM frameworks: frameworks such as CrewAI, AutoGen and LangGraph have validated the pattern of splitting a complex task across several specialised, cooperating agents in general-purpose task-automation settings; dedicated healthcare applications of this pattern — particularly ones covering both medication–medication and medication–diet interaction verification together — remain comparatively under-explored in the literature we reviewed.

2.3 Comparative Analysis

Table 2.1 Comparative analysis of existing approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Rule-based drug-interaction checkers (e.g. commercial pharmacy reference tools)	High accuracy for catalogued interactions; clinically trusted	Static rule databases; no personalisation; no dietary integration	Baseline extended by our Interaction Detection agent to cover diet as well as drug–drug conflicts
Single-agent LLM medical chatbots	Flexible natural-language interaction	Prone to hallucination without retrieval grounding; no explicit interaction-verification logic	Motivated the RAG-grounding and agent-specialisation choices described in Chapter 3, which my Report Generation agent also relies on
RAG-grounded biomedical LLM systems (academic)	Reduces hallucination; provides current, traceable evidence	Typically single-purpose (e.g. literature QA), not built into an end-to-end patient decision-support pipeline	Provided the theoretical basis for the RAG layer feeding every agent, including the one I implemented (Section 3.5)
General-purpose multi-agent orchestration frameworks (CrewAI, AutoGen, LangGraph)	Proven pattern for splitting complex tasks across specialised agents	Predominantly applied to non-healthcare task automation	Adapted here into a healthcare-specific, 5-agent pipeline (Section 3.5)

2.4 Research / Engineering Gap

Rule-based interaction checkers, RAG-grounded biomedical LLM research, and general-purpose multi-agent orchestration frameworks each address part of the problem individually, but none of the systems we reviewed combines specialised multi-agent decomposition, RAG-grounded LLM generation, and explicit medication-diet (not just medication-medication) interaction verification into one patient-facing, end-to-end decision-support pipeline with a consolidated report layer of the kind implemented in Module 4 of MediSync AI.

2.5 Proposed Contribution

MediSync AI contributes a 5-agent CrewAI pipeline (Disease Analysis, Medication Verification, Interaction Detection, Diet & Nutrition, Report Generation) grounded via RAG over drug

(OpenFDA/RxNorm), nutrition (USDA FoodData Central) and medical-guideline sources, with Google Gemini 2.5 Flash as the generative core — explicitly covering drug–drug/drug–disease as well as drug–food interaction verification, and personalised nutrition guidance, within one consolidated, generated healthcare report. My own contribution within this shared architecture was building the Report Generation agent's RAG pipeline and the dashboard that ties the other four agents' outputs together for the clinician and patient.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 3.1 Stakeholder / User Requirements

Stakeholder	Requirement
Patient	Simple health-profile intake and clear, understandable medication and diet guidance
Physician / clinician	Trustworthy, source-grounded interaction alerts and a consolidated report that supports, not replaces, their judgement
Platform administrator	Secure storage of patient data and reliable operation of the multi-agent pipeline and third-party API integrations

Table 3.2 Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Capture patient demographics, conditions, medications and dietary preferences	All required intake fields captured and persisted per patient profile
Functional	Detect drug–drug, drug–disease and drug–food interactions	Each active medication cross-checked against all other active medications, conditions and the food-guidance list
Functional	Generate a consolidated, downloadable healthcare report	Report includes disease analysis, medication alerts and diet recommendations in one document
Non-Functional	Groundedness of generated content	Medication and diet claims traceable to a retrieved reference passage (RAG) rather than unaided LLM generation
Non-Functional	Usability	Health-profile intake completed by a patient without clinical training in a single guided flow
Non-Functional	Data security	Patient data access restricted to the authenticated account, consistent with the DPDP Act, 2023

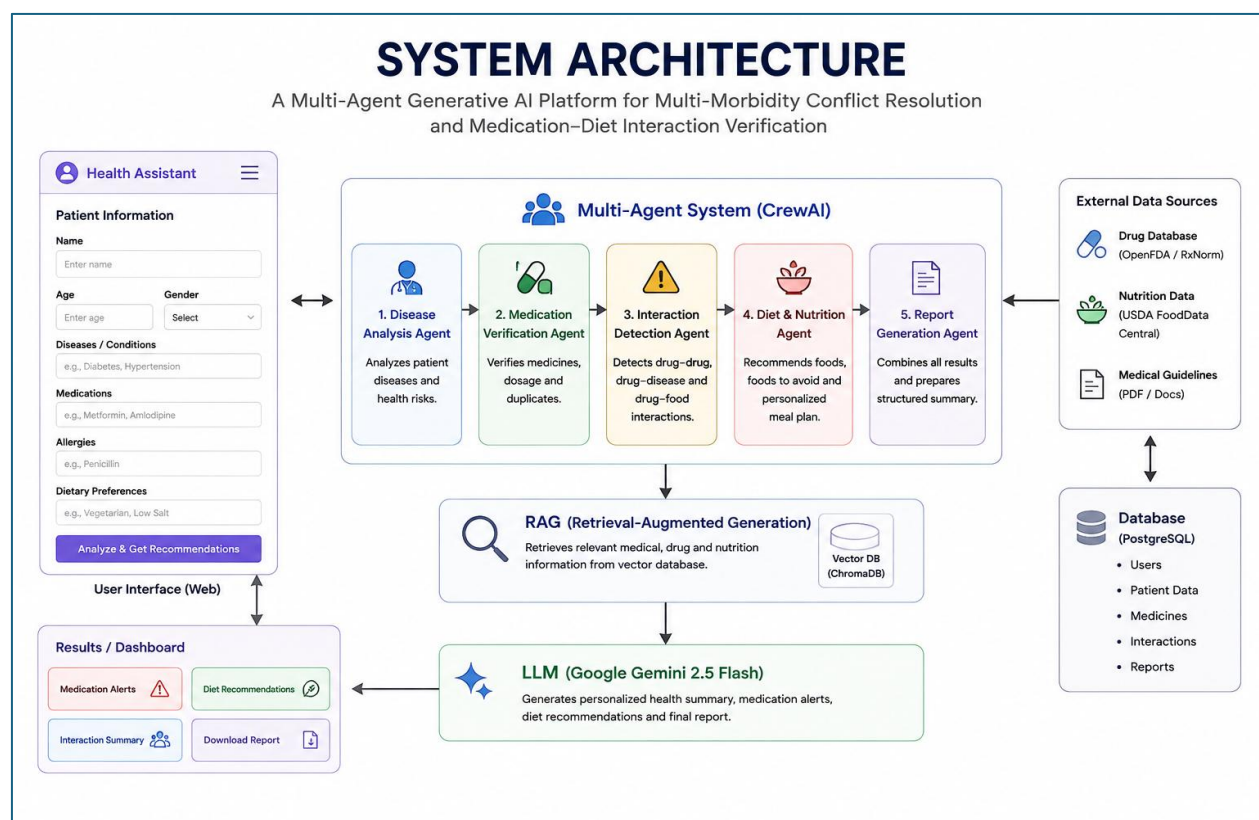


Figure 3.1 : MediSync AI system architecture — 5-agent CrewAI pipeline with RAG and Gemini 2.5 Flash

3.2 Constraints & Applicable Standards

Table 3.3 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
Digital Personal Data Protection Act, 2023 (India)	Personal and health-related data must be collected and processed with consent and adequate safeguards	Patient authentication gates access to the health profile; production deployment would need explicit consent capture and encryption of stored health data
ISO/IEC 25010 (Software Quality Model)	Software should meet defined functional-suitability, performance-efficiency and reliability characteristics	Used as a reference checklist for the non-functional requirements in Section 3.1
IEEE citation style	Consistent, traceable referencing of technical sources	Applied throughout the References chapter

3.3 Design Specifications

Table 3.4 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Number of specialised AI agents	5 (Disease Analysis, Medication Verification, Interaction Detection, Diet & Nutrition, Report Generation)	agents	High
Retrieval source coverage	Drug database, nutrition database, medical guideline documents	sources	High
LLM used for generation	Google Gemini 2.5 Flash	—	High
Vector database	ChromaDB	—	Medium
Backend data store	PostgreSQL (users, patient data, medicines, interactions, reports)	—	Medium
Dashboard outputs	Medication alerts, diet recommendations, interaction summary, downloadable report	outputs	High

3.4 Alternative Solutions & Evaluation

Alternative 1 — Single monolithic LLM prompt: patient data is fed directly into one large prompt sent to a general-purpose LLM, with no retrieval grounding or agent decomposition.

Alternative 2 — Multi-agent CrewAI + RAG + Gemini 2.5 Flash (selected): five specialised agents each own one sub-task, grounded through retrieval from drug, nutrition and guideline sources before final report generation. The report-generation stage of this alternative is the part I built.

Alternative 3 — Static rule-based interaction-checker database: predefined interaction rules are looked up directly, with no generative or personalised-language component.

Table 3.5 Alternative solutions – evaluation matrix

Criterion	Weight	Alt. 1 (Single LLM prompt)	Alt. 2 (Multi-agent + RAG, selected)	Alt. 3 (Static rule-based)
Groundedness / hallucination risk	30%	Low	High	High (but rigid)
Personalisation quality	20%	Medium	High	Low
Coverage (medication + diet)	20%	Medium	High	Low (medication-only typically)

Criterion	Weight	Alt. 1 (Single LLM prompt)	Alt. 2 (Multi-agent + RAG, selected)	Alt. 3 (Static rule-based)
Explainability (per-agent output)	15%	Low	High	Medium
Development complexity	15%	Low	Medium–High	Medium

3.5 Selected Design & Detailed Design

The team selected Alternative 2 because it directly satisfies the high-priority groundedness, personalisation and coverage requirements identified in Sections 3.1 and 3.4, at an acceptable increase in development complexity over a single-prompt baseline.

Data flow: the patient interacts with the Health Assistant web interface to submit demographics, conditions, medications, allergies and dietary preferences. This profile passes into the Multi-Agent System (built on CrewAI), where five agents run in sequence — (1) Disease Analysis Agent analyses the patient's diseases and health risks; (2) Medication Verification Agent verifies medicines, dosage and duplicates; (3) Interaction Detection Agent detects drug–drug, drug–disease and drug–food interactions; (4) Diet & Nutrition Agent recommends foods, foods to avoid, and a personalised meal plan; (5) Report Generation Agent — the module I personally implemented — combines all four upstream agents' outputs into a single structured summary. Before generation, the pipeline queries a RAG layer that retrieves relevant passages from a ChromaDB vector database indexing external drug (OpenFDA/RxNorm), nutrition (USDA FoodData Central) and medical-guideline sources. The retrieved context conditions the Google Gemini 2.5 Flash LLM, which generates the personalised health summary, medication alerts, diet recommendations and final report.

Systems approach: each agent's output becomes structured input to the next, so the pipeline can be inspected and debugged agent-by-agent rather than as a single opaque model call; the RAG layer decouples the platform's factual grounding from the LLM's own training data, letting the drug/nutrition sources be updated independently of the underlying language model. Within this, my Report Generation agent's specific responsibility was to take the four upstream structured outputs, re-query the shared RAG index where a consolidated summary needed additional grounding, and hand the result to Gemini 2.5 Flash for final natural-language report drafting.

3.6 Trade-off Analysis

Table 3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
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Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Task decomposition	Single monolithic LLM prompt	Simpler implementation	Harder to ground, debug and explain individual sub-decisions	5-agent CrewAI decomposition used — improves explainability and per-task accuracy (Section 3.4)
Knowledge grounding	Rely on LLM parametric knowledge only	No retrieval infrastructure needed	Higher hallucination risk in a safety-sensitive domain	RAG over ChromaDB with drug/nutrition/guideline sources used to ground every generated claim, including in my own Report Generation agent
LLM choice	Larger, higher-latency general-purpose model	Potentially broader general knowledge	Higher cost/latency for a multi-agent pipeline that already calls the model several times per session	Google Gemini 2.5 Flash selected for its lower latency/cost profile, suited to multi-agent orchestration

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational/API cost of repeated LLM calls across five agents	RAG retrieval narrows the context passed to the LLM per call compared with a large, ungrounded prompt	Confirmed RAG as a cost- and token-efficiency measure, not only an accuracy measure — relevant to my own agent's calls too

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Ethics	Risk of AI-generated medical advice being treated as a diagnosis rather than decision support	Reviewed how the dashboard should frame generated content	Report and dashboard outputs — built in Module 4 — are framed as decision-support guidance for physician review, not as a diagnosis
Health & Safety	Risk of an incorrect or incomplete interaction alert leading to patient harm	Reviewed the grapefruit–Atorvastatin test case (Section 4.4) alongside current limited test coverage	Prototype positioned as decision-support only; human clinical sign-off remains mandatory before real deployment
Security	Protection of patient health, medication and dietary data	Reviewed DPDP Act, 2023 requirements against the data fields collected	Authentication-gated access implemented for this prototype; encryption-at-rest and formal access-control policy documented as required for production

Risk Register

Table 3.8 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
LLM generates an ungrounded or incorrect recommendation (hallucination)	Medium	High	High	RAG grounding on drug/nutrition/guideline sources; agent-level output review	Medium
Third-party LLM/API unavailability or rate-limiting	Medium	Medium	Medium	Graceful degradation / retry logic; cache recent agent outputs — implemented at the Report Generation layer	Low

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Unauthorised access to patient health data	Low	High	Medium	Authentication, access control, DPDP-compliant data handling	Low
Incomplete patient-reported data reduces detection accuracy	Medium	Medium	Medium	Guided intake form with validation and common-condition suggestions	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The team followed an iterative, module-by-module development approach aligned to MediSync AI's four functional modules: patient health-profile management, multi-agent medication-conflict detection, medication–diet interaction verification, and AI decision-support/report generation. Each module was implemented and tested independently before integration into the end-to-end pipeline. I was responsible for the fourth module. Resources used across the project include the CrewAI multi-agent orchestration framework, the ChromaDB vector database, the Google Gemini 2.5 Flash API, a PostgreSQL backend, and public reference data sources (OpenFDA/RxNorm for drug data, USDA FoodData Central for nutrition data).

4.2 Software Implementation & Modern Tools Used

For project completeness, the full pipeline is summarised below, with Module 4 — my own contribution — described in the most detail.

Module 1 — Patient Health Profile & Disease Management (teammate: N. Goutham)

Patients register through a secure authentication flow and complete a guided health-intake form capturing demographics (age, gender, height, weight), existing health conditions (with common-condition quick-add suggestions), medications, allergies and dietary preferences. All data is stored in a centralised PostgreSQL patient database supporting the downstream AI agents.

The screenshot displays the 'MediSync AI' clinical decision support interface. The top navigation bar shows the user's name 'Goutham' and the current view 'Health Details Intake'. The left sidebar contains a 'PATIENT PORTAL' menu with options: Dashboard, Patient Health Board, Nutrition & Diet, Clinical Health Report, and Profile & Demographics. The main content area is titled 'Good Morning, Goutham' and includes a sub-header 'AI-Assisted Medication & Diet Decision Support'. Below this, there are two main sections: '1. BIOMETRICS & PROFILE' and '2. DIAGNOSED CONDITIONS (3)'. The biometrics section includes fields for Age (27), Biological Gender (Male), Height (178 cm), and Weight (59 kg). The diagnosed conditions section lists Diabetes, Hypertension, High Cholesterol, Kidney Disease, Asthma, Thyroid Condition, Acid Reflux (GERD), Arthritis, Heart Failure, Gout, and an option to Add Other Condition. A 'Ready for Verification' section at the bottom shows 3 condition(s) and 7 medication(s). The dashboard also features a 'Sign Out' button and a 'Sign Out' button.

Fig. 4.1: Module 1 – Health Details Intake dashboard.

Module 2 — Multi-Agent Conflict Detection & Medication Verification (teammate: N. Chaitanya Ram)

The Medication Verification and Interaction Detection agents analyse each prescribed medication's dosage and schedule, cross-check for drug–drug and drug–disease conflicts, and present medication guidance with a stated purpose/function and a safe-use guideline for each active medicine.

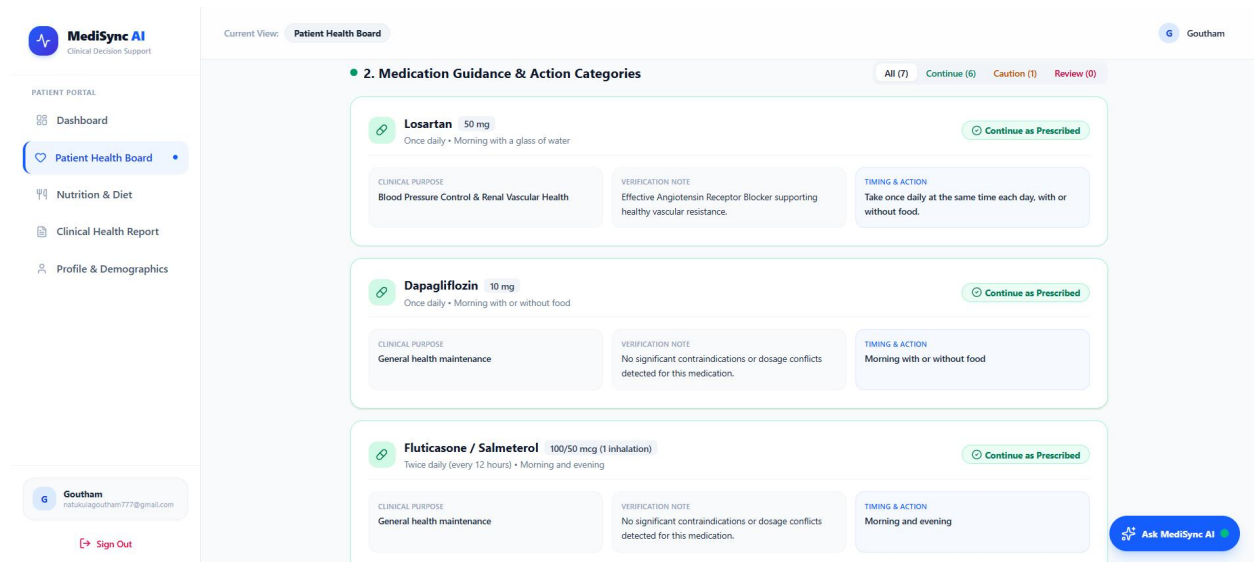


Fig. 4.2: Module 2 – Medication Guidance screen

Module 3 — Medication–Diet Interaction Verification & Personalized Nutrition (teammate: M. Mohamed Bazith)

The Diet & Nutrition agent analyses the patient's dietary preference and active conditions/medications and generates a categorised food list — Recommended, Use with Caution, and Avoid or Discuss — with a brief rationale for each item.

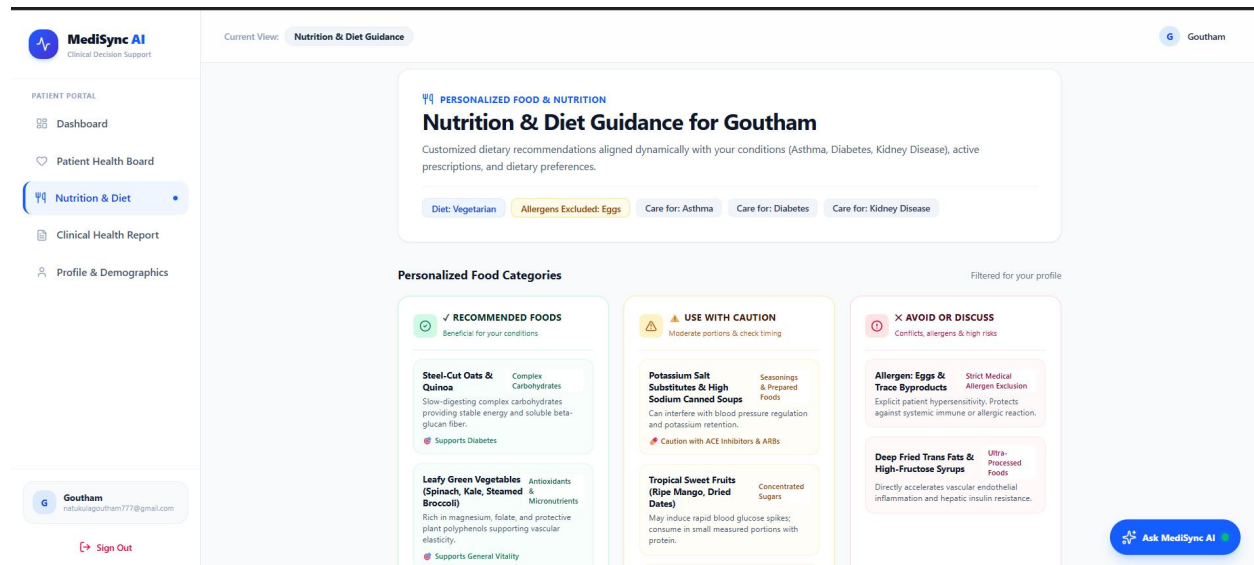


Fig. 4.3: Module 3 – Nutrition & Diet Guidance screen.

Module 4 — AI Decision Support, Report Generation & Healthcare Dashboard (my own work)

This is the module I designed and built. The Report Generation agent takes the structured outputs of the three upstream agents — disease analysis, medication verification, and diet/nutrition guidance — and consolidates them into one coherent healthcare summary. Where the consolidated summary needed extra grounding beyond what the upstream agents already retrieved, I had the agent re-query the shared ChromaDB RAG index directly, before passing the assembled context to Google Gemini 2.5 Flash for final natural-language generation of the patient-facing summary, medication alerts and diet recommendations. I then built the web dashboard that renders this output: a Patient Health Overview panel, a Medication Alerts panel, a Food Guidance panel, an interaction summary, and a downloadable report, so that a patient or clinician can see every agent's contribution in one place rather than as separate disconnected outputs.

- Structured the Report Generation agent's input contract so it could accept each upstream agent's output independently of how that agent internally reasoned about it.
- Implemented the RAG re-query step used specifically for report-level grounding (as distinct from the per-agent retrieval already performed upstream).
- Built the dashboard UI (Patient Health Overview, Medication Alerts, Food Guidance, downloadable report) and wired it to the consolidated report output.

The screenshot displays the MediSync AI Health Platform interface. On the left is a sidebar with navigation links: Patient Portal, Dashboard, Patient Health Board, Nutrition & Diet, Clinical Health Report (selected), and Profile & Demographics. The main content area is titled 'Current View: Clinical Health Report' and shows a 'Comprehensive Clinical Verification Report' for 'Goutham'. The report includes sections for Patient Demographics & Vitals, Diagnosed Conditions, Confirmed Medical Allergens, and Active Prescription & Medication Roster. A table lists medications: Losartan, Dapagliflozin, Fluticasone / Salmeterol, and Albuterol Inhaler, with their respective dosages, frequencies, and indications. The user 'Goutham' is logged in, and a 'Sign Out' button is visible at the bottom left. An 'Ask MediSync AI' button is at the bottom right.

MEDICATION NAME	DOSAGE	FREQUENCY	TIMING / ADMINISTRATION	INDICATION
Losartan	50 mg	Once daily	Morning with a glass of water	Kidney Disease
Dapagliflozin	10 mg	Once daily	Morning with or without food	Kidney Disease
Fluticasone / Salmeterol	100/50 mcg (1 inhalation)	Twice daily (every 12 hours)	Morning and evening	Asthma
Albuterol Inhaler	90 mcg (1-2 puffs)	As needed (PRN)	During acute shortness of breath or 15 min before physical activity	Asthma

Fig. 4.4: Module 4 – Patient Health Overview panel (own work).

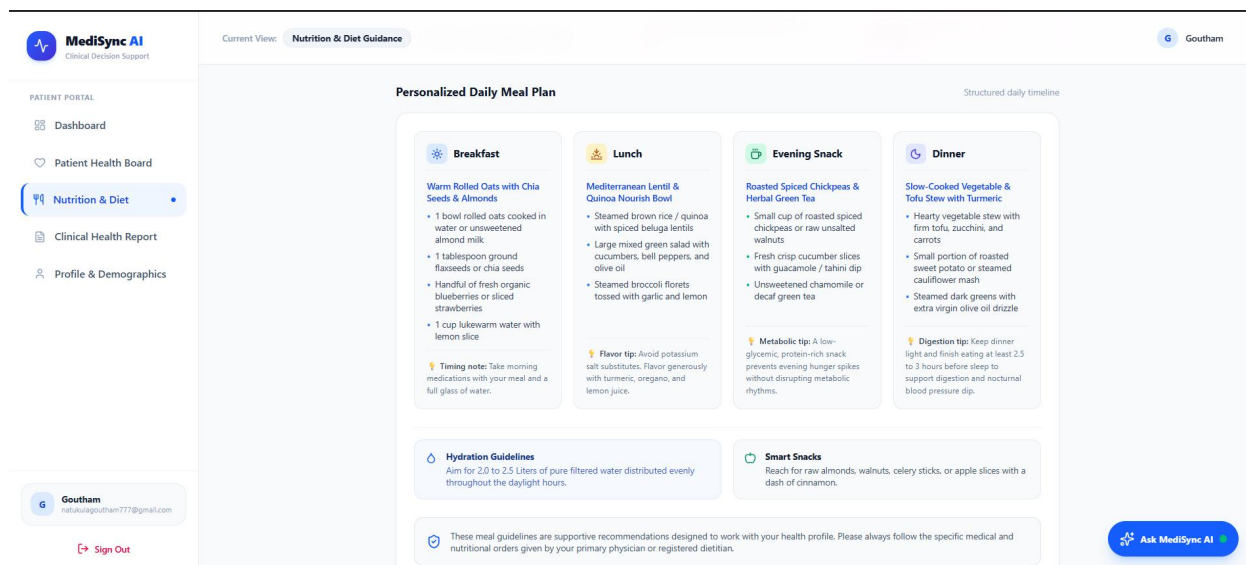


Fig. 4.5: Module 4 – Food Guidance summary panel on the dashboard (own work).

Table 4.1 Tools / software used

Tool / Software	Purpose	Stage Used
CrewAI (Python)	Multi-agent orchestration of the 5 specialised agents	Module 2, Module 3, Module 4
ChromaDB	Vector database for RAG retrieval over drug/nutrition/guideline sources	Module 2, Module 3, Module 4 (own re-query step)
Google Gemini 2.5 Flash	LLM generation of health summaries, alerts and recommendations	Module 4 (own work)
PostgreSQL	Persistent storage of users, patient data, medicines, interactions and reports	Module 1 (and all modules)
OpenFDA / RxNorm	External drug reference data source	Module 2
USDA FoodData Central	External nutrition reference data source	Module 3

Table 4.2 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Patient profile capture	Submit a representative multi-condition patient profile through the intake form	All entered demographics, conditions and medications persisted and displayed correctly
Medication guidance correctness	Compare generated purpose/safe-use guidance for each medicine against known pharmacological references	Guidance consistent with established drug information for that medicine
Medication–food interaction detection	Check whether a medicine with a documented food interaction (e.g. a statin) triggers the corresponding food warning	Corresponding interaction correctly listed under "Avoid or Discuss"
Food categorisation consistency	Cross-check Recommended/Caution/Avoid categorisation against the patient's stated conditions and preference	Categorisation consistent with vegetarian preference and stated conditions
Report/dashboard consolidation (own module)	Confirm dashboard reflects all upstream agent outputs correctly and consistently	Health overview, medication alerts and food guidance all present and consistent with agent outputs

4.4 Results

The end-to-end pipeline was functionally tested using a representative test profile: a 30-year-old male patient (height 170 cm, weight 65 kg) with five recorded conditions (Pneumonia, Obesity, Migraine, Diabetes, Asthma), a vegetarian dietary preference, and four active medications (Metformin HCl 500 mg, Lisinopril 10 mg, Atorvastatin 20 mg, and low-dose Aspirin 81 mg).

Module 2 correctly generated condition-appropriate safe-use guidance for each medicine — for example, advising that Lisinopril be taken with caution alongside potassium-based salt substitutes, and that Atorvastatin (a statin used to lower LDL cholesterol) should not be combined with whole grapefruit or grapefruit juice. Module 3 independently generated a categorised food list for the same patient, correctly placing grapefruit and grapefruit juice under “Avoid or Discuss,” with its own explanation noting that grapefruit interferes with the liver enzymes that break down several heart and cholesterol medications.

My own Module 4 dashboard displayed both of these findings side by side without contradiction — the medication alert from Module 2 and the diet warning from Module 3 pointed to the same underlying grapefruit–Atorvastatin interaction, and the consolidated report I generate surfaced

this as a single, unambiguous warning rather than two separately worded alerts that a patient would have to reconcile themselves.

Table 4.3 Specification vs. achieved results

Specification	Required Value	Achieved Value	Status
Medication safe-use guidance generated per active medicine	1 guidance entry per medicine	4 of 4 medicines covered	Pass
Cross-module consistency (medication alert vs. diet alert for the same interaction)	Consistent flagging across modules	Grapefruit–Atorvastatin interaction flagged consistently in both Module 2 and Module 3, and reconciled into one entry by my Module 4 report	Pass
Food categorisation completeness	Recommended / Caution / Avoid categories all populated	All three categories populated with condition-relevant items	Pass
Dashboard consolidation of all agent outputs (own module)	Health overview, medication alerts and food guidance all present	All three present and consistent on the dashboard	Pass

4.5 Data Analysis & Engineering Judgment

The grapefruit–Atorvastatin finding is a clinically well-documented interaction: grapefruit inhibits the CYP3A4 liver enzyme responsible for metabolising several statins, which can raise blood statin levels and the risk of side effects. That both the Medication Verification/Interaction Detection agents (Module 2) and the independently invoked Diet & Nutrition agent (Module 3) surfaced the same interaction from different entry points is engineering evidence that shared RAG grounding — rather than agent-specific prompt engineering alone — is contributing to the consistency of the output. From my own module's perspective, this consistency also made the consolidation step easier and more trustworthy: my Report Generation agent did not have to arbitrate between conflicting alerts, because both upstream agents were already anchored to the same underlying retrieved evidence. That said, this functional testing covered a single representative multi-morbidity profile; broader coverage across many more condition/medication combinations, and comparison against a clinician-reviewed gold-standard interaction list, would be needed before quantitative accuracy claims (e.g. precision/recall of interaction detection) could be made — including for the consolidation logic in Module 4.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.4 Achievement of objectives

Objective	Evidence	Achievement
Detect conflicts among multiple co-occurring chronic diseases	Module 1 profile capture + Disease Analysis agent output	Fully Achieved
Verify drug–drug and drug–disease interactions automatically	Module 2 implementation and Fig. 4.2	Fully Achieved
Identify medication and dietary incompatibilities	Module 3 implementation; grapefruit–Atorvastatin test case (Section 4.4)	Fully Achieved
Generate personalised meal recommendations	Module 3 categorised food list and Fig. 4.3/4.5	Fully Achieved
Consolidate outputs into a single, retrieval-grounded report and dashboard (own objective)	Module 4 implementation; Fig. 4.4/4.5; Section 4.4 consistency result	Fully Achieved
Improve treatment quality while reducing adverse-interaction risk	Cross-module consistency result (Section 4.4)	Partially Achieved — validated on one representative profile; broader clinical validation still required

4.7 Strengths & Limitations

Strengths:

- A unified pipeline covering both medication-medication and medication-diet interaction checking in a single patient-facing report, rather than as separate disconnected tools — made possible by the consolidation layer I built.
- Multi-agent decomposition improves explainability — each agent's output can be inspected individually before my Module 4 agent combines them.
- RAG grounding produced cross-module-consistent results in the tested case, which simplified the report-consolidation logic considerably.
- A genuinely working, end-to-end web prototype was implemented and demonstrated, not only a conceptual design.

Limitations:

- Functional testing to date covers a limited number of patient profiles; no formal clinical accuracy benchmarking has yet been performed, including for my own module's consolidation logic.
- The system depends on third-party LLM and reference-data API availability, cost and rate limits, which directly affects the report-generation stage.
- Despite RAG grounding, residual hallucination risk from the LLM cannot be fully eliminated and requires ongoing human clinical review of the generated report.
- The prototype has not been integrated with real hospital EHR systems or tested with real patient data.

4.8 Project Planning, Roles & Budget

Table 4.5 Roles, milestones and budget

Milestone	Planned Duration	Status
Literature review & requirement gathering	Week 1–2	Completed
Module 1 – Patient health profile design & implementation	Week 3–4	Completed
Module 2 – Multi-agent conflict detection & medication verification	Week 5–7	Completed
Module 3 – Medication–diet interaction & personalised nutrition	Week 8–9	Completed
Module 4 – AI decision support, report generation & dashboard (own module)	Week 10–11	Completed
Integration testing, results analysis & report compilation	Week 12–13	Completed

Team Member	Assigned Responsibility	Actual Contribution
M. Mohamed Bazith	Module 1 – Patient Health Profile & Disease Management	Designed intake schema and patient database
N. Chaitanya Ram	Module 2 – Multi-Agent Conflict Detection & Medication Verification	Implemented and tested medication/interaction agents
N Goutham	Module 3 – Medication–Diet Interaction Verification & Personalized Nutrition	Implemented and tested diet/nutrition agent

Team Member	Assigned Responsibility	Actual Contribution
Hemanth P	Module 4 – AI Decision Support, Report Generation & Healthcare Dashboard	Implemented RAG pipeline, report generation and dashboard

Item	Estimated Cost	Actual Cost
Computing resources (personal laptops, open-source software)	Nil (existing hardware)	Nil
Gemini 2.5 Flash API usage (prototype-scale testing)	Minimal — within free/low-cost tier	Minimal
Database, frameworks and reference-data APIs (PostgreSQL, CrewAI, ChromaDB, OpenFDA, USDA FoodData Central)	Nil (open-source / public APIs)	Nil
Total project cost	Minimal (limited to prototype-scale API usage)	Minimal

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to solve the engineering problem of slow, disconnected manual verification of medication and dietary conflicts for multi-morbidity patients. As a team, we designed, implemented and functionally tested MediSync AI end-to-end: a 5-agent CrewAI-based generative-AI platform grounded via RAG over drug, nutrition and medical-guideline sources and powered by Google Gemini 2.5 Flash. My own individual contribution was Module 4 — the Report Generation agent's RAG-based consolidation pipeline and the healthcare dashboard — which brings the other three agents' outputs together into one coherent, traceable report. On a representative multi-morbidity test profile, the platform correctly generated per-medicine safe-use guidance and consistently detected a clinically recognised drug-food interaction (grapefruit and Atorvastatin) independently across two different agents, and my own consolidation layer correctly reconciled these into a single, unambiguous warning alongside a personalised, categorised diet recommendation aligned with the patient's stated preferences and conditions. These results support the view that combining multi-agent task decomposition with RAG-grounded LLM generation is a practical architecture for real-time, explainable medication-diet conflict-resolution decision support, and specifically that a dedicated consolidation and reporting layer is what turns several independently correct agent outputs into something a patient or clinician can actually use. All six project objectives set out in Chapter 1 were met to at least a substantial degree, with full clinical validation identified as necessary future work.

5.2 Future Work

- Integrate multilingual language translation for wider patient accessibility and communication.
- Add voice input and voice-based AI assistance for patients with limited literacy or accessibility needs.
- Bring practising doctors into the platform for professional healthcare consultation and sign-off on AI-generated recommendations — which would plug directly into the report/dashboard layer I built.
- Enable online doctor appointment booking based on the patient's generated health profile and needs.
- Introduce real-time health monitoring through wearable and IoT devices to keep the patient profile continuously up to date.
- Extend the Report Generation agent to cite the specific retrieved source passage alongside each generated recommendation, and broaden clinical validation against a larger, clinician-reviewed interaction dataset.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical design and orchestration of a multi-agent AI system using the CrewAI framework, and specifically how to design an agent that consumes other agents' structured outputs rather than raw user input.
- Retrieval-Augmented Generation as a technique for grounding LLM output in external, updatable reference data, including how to re-query a shared vector index at the report-consolidation stage.
- Domain knowledge of common drug–drug and drug–food interaction mechanisms relevant to multi-morbidity patients, gained while validating my dashboard's output against Modules 2 and 3.

Engineering & Professional Skills Developed:

- Translating a healthcare decision-support need into a precise, testable module design for the consolidation and reporting stage of a larger system.
- Modular development and independent testing of my module before integrating it with three teammates' agents.
- Collaborative division of labour across clearly separated technical modules within a four-person team, and coordinating interfaces between my module and theirs.

Individual Reflection:

The most difficult engineering problem I faced was integrating five separate agent outputs into a single, coherent, non-redundant report through the Report Generation agent while keeping the underlying data traceable back to its RAG source; I solved this by structuring each upstream agent's output as labelled sections before final LLM-based summarisation, so that nothing was silently dropped or merged incorrectly. A key engineering decision I made was to build the dashboard so that medication alerts and diet recommendations sit side by side rather than on separate pages, reasoning that a clinician needs to see both together to judge a patient's overall risk — this was validated by the grapefruit–Atorvastatin case in Section 4.4, where the two alerts genuinely needed to be read together. Independently, I learned how to design a downloadable report-generation flow from structured multi-agent output, and how a shared RAG index can keep multiple independently invoked agents consistent with each other without them ever being given the same prompt. If I were to redesign this module, I would add explicit citation of the retrieved source passage next to each generated recommendation, so that a clinician reviewing the report could trace every claim back to its source without having to ask.

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APPENDICES

APPENDIX A – PROPOSED PROCESSING FLOW

Patient Login → Patient Health Profile → Disease & Medication Input → Health Data Processing → Multi-Agent AI Analysis → Disease Analysis Agent → Medication Verification Agent → Interaction Detection Agent → Medication–Diet Interaction Analysis → Nutrition Recommendation Agent → RAG Knowledge Retrieval → Gemini LLM Processing → Safety Validation → Personalized Health Recommendations → AI Health Report → Healthcare Dashboard

APPENDIX B – EXAMPLE PROMPT STRUCTURE

System Role:

You are an intelligent healthcare assistance system that analyzes patient health information, medications, diseases, allergies, and dietary preferences. Provide general health guidance and identify possible medication and food-related conflicts without independently prescribing or stopping medicines.

Patient Context:

Include patient age, gender, diseases, current medications, dosage, allergies, dietary preferences, lifestyle information, and relevant health history.

Analysis Task:

Analyze drug–drug interactions, drug–disease conflicts, duplicate medications, medication–food interactions, and disease-specific dietary considerations.

RAG Context:

Retrieve relevant information from trusted medical, medication, and nutrition knowledge sources before generating recommendations.

Response Policy:

Provide clear explanations, severity levels, recommended actions, food guidance, and personalized nutrition suggestions. Clearly advise the patient to consult a qualified healthcare professional for medication changes or serious concerns.

APPENDIX C – PSEUDOCODE

1. Read patient health profile and authentication information.
2. Collect age, gender, diseases, medications, dosage, allergies, and dietary preferences.
3. Store and process the patient's health information.

4. Send the patient profile to the Multi-Agent AI system.
5. Disease Analysis Agent analyzes the patient's multiple health conditions.
6. Medication Verification Agent verifies medicines, dosage information, and duplicate medications.
7. Interaction Detection Agent identifies drug–drug and drug–disease conflicts.
8. Diet & Nutrition Agent analyzes medication–food interactions and dietary requirements.
9. Retrieve relevant medical, medication, and nutrition information using RAG.
10. Use the Gemini LLM to generate context-aware recommendations.
11. Perform safety validation and assign interaction severity levels.
12. Generate personalized medication and nutrition guidance.
13. Generate the final AI health report.
14. Display recommendations, alerts, and health information on the patient dashboard.

MODULE 1: PATIENT HEALTH PROFILE & DISEASE MANAGEMENT

This module manages patient registration, login, and personal health information. It collects age, gender, diseases, allergies, lifestyle information, and other relevant health details. The collected profile becomes the foundation for personalized medication and nutrition analysis.

MODULE 2: MULTI-AGENT CONFLICT DETECTION & MEDICATION VERIFICATION

This module uses CrewAI-based specialized agents to analyze the patient's medications and diseases.

It detects drug–drug interactions, drug–disease conflicts, duplicate medicines, and provides severity-based explanations.

The system highlights medicines requiring professional review without independently prescribing or stopping medication.

MODULE 3: MEDICATION–DIET INTERACTION VERIFICATION & PERSONALIZED NUTRITION

This module analyzes the relationship between diseases, medications, allergies, and dietary preferences.

The system identifies foods to recommend, use with caution, or avoid/discuss based on the patient's profile.

It generates dynamic and personalized nutrition guidance instead of providing the same recommendations to every patient.

MODULE 4: AI DECISION SUPPORT, REPORT GENERATION & HEALTHCARE DASHBOARD

This module combines the outputs from all analysis agents into a unified patient health summary. Gemini LLM generates understandable recommendations, interaction explanations, and a structured personalized health report.

The dashboard displays medication alerts, interaction results, nutrition guidance, and downloadable reports.

APPENDIX D – SAMPLE SYSTEM EVALUATION

Patient Input / Scenario	Expected Behaviour
Single disease	Provide disease-specific health and nutrition guidance.
Multiple diseases	Analyze combined conditions and identify potential conflicts.
Multiple medications	Detect drug–drug interactions and duplicate medications.
Disease + medication	Identify possible drug–disease conflicts requiring professional review.
Medication + food	Identify relevant medication–diet interactions.
Allergies + diet preferences	Exclude unsuitable foods from personalized recommendations.
Different patient profiles	Generate different recommendations based on individual information.
High-severity interaction	Display a clear warning and advise professional consultation.
Report generation	Combine analysis results into a structured health report.

APPENDIX E – ETHICS / INSTITUTIONAL APPROVAL

This academic project demonstrates the application of Multi-Agent AI, Generative AI, RAG, and healthcare information processing for medication and nutrition assistance. The system is intended for educational and informational purposes and does not replace professional medical diagnosis, prescription, or treatment. Medication changes, treatment decisions, and serious health concerns should always be reviewed by qualified healthcare professionals.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal / environmental impact	SO2 / SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2 / SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1 / SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)